

An Exact Cooperation Formula for Introspection Dynamics in the Heterogeneous Public Goods Game

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Abstract

We study introspection dynamics on the heterogeneous public goods game, where N players may differ in their contributions α_i , public goods multipliers r_i , and choice intensities β_i . We prove that the counterfactual payoff difference a player evaluates when considering a strategy switch is independent of all other players' current actions, a consequence of the linear payoff structure. This state-independence implies that the introspection Markov chain decomposes as a random-scan product of N independent two-state chains, one per player. The stationary distribution is therefore a product measure, and the long-run cooperation probability admits the exact closed form

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1 - 2\mu}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu \right],$$

with no asymptotic approximation. Several structural consequences follow immediately: a player-specific cooperation threshold at $r_i = N$, neutrality under zero choice intensity, and the sign of each player's sensitivity to their own parameters.

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1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed equally, creating a tension between individual incentive and collective benefit [4]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences.

A natural learning rule for such settings is *introspection dynamics*, introduced by Couto, Giaimo, and Hilbe [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played the alternative action (keeping all other players fixed), and switches with a probability governed by this counterfactual difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

Couto and Pal [2] extended introspection dynamics to multiplayer asymmetric games and derived a formula for the stationary distribution that can be evaluated numerically for any game. In their general framework the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N ; no closed form is available in general.

In this paper we show that the specific payoff structure of the heterogeneous public goods game admits an exact closed-form analysis. The key observation is that the counterfactual payoff difference $\Delta\pi_i$ a player evaluates depends only on their own current action and individual parameters, not on the actions of others. This is a consequence of the *linearity* of the public goods payoff in contributions. As a result, the introspection chain decomposes exactly as a product of N independent two-state chains (one per player), the stationary distribution is a product measure, and the cooperation probability p_C^{intro} admits an explicit closed form that scales as $O(N)$ rather than $O(2^N)$.

The paper is organised as follows. Section 2 introduces the model. Section 3 establishes the payoff identity. Section 4 proves the product decomposition and derives the exact cooperation formula. Section 5 records several structural corollaries. Section 6 concludes.

2 Setup and notation

Consider a well-mixed population of N ordered players with contributions $\alpha_1, \dots, \alpha_N > 0$, player-specific public goods multipliers $r_1, \dots, r_N > 1$, player-specific choice intensities $\beta_1, \dots, \beta_N > 0$, and two actions $\{C, D\}$ coded as $\{1, 0\}$. The state space is $S = \{0, 1\}^N$ with $|S| = 2^N$.

Payoff. The payoff to player i in state $\mathbf{a} = (a_1, \dots, a_N)$ is

$$\pi_i(\mathbf{a}) = \frac{r_i}{N} \sum_{j=1}^N \alpha_j a_j - \alpha_i a_i. \quad (1)$$

We write $C(\mathbf{a}) = \sum_j \alpha_j a_j$ for the total contribution.

Fermi function. Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$, satisfying $\phi_i(x) + \phi_i(-x) = 1$ and $\phi_i(0) = \frac{1}{2}$.

Mutation. All dynamics carry a uniform mutation rate $\mu \in (0, \frac{1}{2})$, ensuring ergodicity of the Markov chain.

Definition 1 (Cooperation probability). *Given an ergodic chain on S with stationary distribution π , the cooperation probability is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N} \sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

Introspection dynamics. At each step, one player i is selected uniformly at random. Player i considers switching from action a_i to the alternative $\bar{a}_i = 1 - a_i$. Writing $\mathbf{a}_{i \rightarrow \bar{a}_i}$ for the state obtained by flipping only coordinate i , the transition probability for the switch is

$$T_{\mathbf{a}, \mathbf{a}_{i \rightarrow \bar{a}_i}}^{\text{intro}} = \frac{1}{N} [\phi_i(\Delta\pi_i)(1 - 2\mu) + \mu], \quad (2)$$

where $\Delta\pi_i := \pi_i(\mathbf{a}) - \pi_i(\mathbf{a}_{i \rightarrow \bar{a}_i})$ is the counterfactual payoff gain from keeping action a_i . Since ϕ_i is strictly decreasing, a larger payoff gain from the current action yields a smaller switching probability.

3 The payoff identity

Lemma 2 (State-independence of counterfactual payoff differences). *For the payoff (1), the counterfactual difference player i evaluates when considering a switch from a_i to $\bar{a}_i$ is*

$$\Delta\pi_i = (2a_i - 1) \alpha_i \left(\frac{r_i}{N} - 1 \right). \quad (3)$$

This depends only on a_i , α_i , r_i , and N ; in particular, it is independent of all other players' actions $\mathbf{a}_{-i}$.

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own contribution changes, so the total contribution becomes $C(\mathbf{a}_{i \rightarrow \bar{a}_i}) = C(\mathbf{a}) + \alpha_i(1 - 2a_i)$. Expanding both payoffs:

$$\begin{aligned} \pi_i(\mathbf{a}) &= \frac{r_i}{N} C(\mathbf{a}) - \alpha_i a_i, \\ \pi_i(\mathbf{a}_{i \rightarrow \bar{a}_i}) &= \frac{r_i}{N} [C(\mathbf{a}) + \alpha_i(1 - 2a_i)] - \alpha_i(1 - a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned} \Delta\pi_i &= -\frac{r_i}{N} \alpha_i(1 - 2a_i) + \alpha_i(1 - 2a_i) \\ &= \alpha_i(1 - 2a_i) \left(1 - \frac{r_i}{N} \right) \\ &= \alpha_i(2a_i - 1) \left(\frac{r_i}{N} - 1 \right). \end{aligned}$$

The term $C(\mathbf{a})$ cancels entirely, so $\Delta\pi_i$ is independent of $\mathbf{a}_{-i}$. □

Remark 3. *The cancellation of $C(\mathbf{a})$ relies on the payoff (1) being linear in contributions. The heterogeneity of r_i does not affect this cancellation: each player's counterfactual comparison depends only on their own parameters.*

4 Product decomposition and exact cooperation probability

A Markov chain on a product space $S = S_1 \times \cdots \times S_N$ is called a *random-scan composition* [3] if each step (i) selects a coordinate i uniformly at random and (ii) updates coordinate i according to a kernel M_i that may depend on coordinate i but not on the remaining coordinates $\mathbf{a}_{-i}$.

Theorem 4 (Product decomposition). *Under introspection dynamics with player-specific β_i and r_i , uniform mutation $\mu \in (0, \frac{1}{2})$, and payoff (1), the Markov chain on $S = \{0, 1\}^N$ is a random-scan product of N independent two-state chains with stationary distributions $\pi_i(C) = p_i$, $\pi_i(D) = 1 - p_i$, where*

$$p_i = \phi_i(\alpha_i(1 - r_i/N))(1 - 2\mu) + \mu. \quad (4)$$

Proof. By Lemma 2, $\Delta\pi_i$ depends on the current state $\mathbf{a}$ only through coordinate a_i . The switching probability (2) therefore depends on $\mathbf{a}$ only through a_i , and the transition for player i defines a 2×2 conditional update kernel M_i (rows indexed by $D = 0, C = 1$, columns by the next action):

$$M_i = \begin{pmatrix} 1 - p_i & p_i \\ q_i & 1 - q_i \end{pmatrix},$$

where p_i is as in (4) and

$$q_i = \phi_i(\alpha_i(r_i/N - 1))(1 - 2\mu) + \mu \quad (C \rightarrow D).$$

Since $\phi_i(x) + \phi_i(-x) = 1$, we have $p_i + q_i = 1$, so $q_i = 1 - p_i$ and both rows of M_i are identical:

$$M_i = \begin{pmatrix} 1 - p_i & p_i \\ 1 - p_i & p_i \end{pmatrix}.$$

Player i 's next action is therefore drawn from $\pi_i = (1 - p_i, p_i)$ regardless of their current action.

Since $[M_i]_{a_i, b_i} = \pi_i(b_i)$ is independent of a_i , the full transition matrix can be written as

$$T_{\mathbf{a}, \mathbf{b}}^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \pi_i(b_i) \prod_{j \neq i} \delta_{a_j, b_j},$$

where $\delta_{a,b}$ is the Kronecker delta. Each term in the sum corresponds to player i being selected: coordinate i is redrawn from π_i , while all other coordinates are unchanged (δ_{a_j, b_j} enforces $a_j = b_j$ for $j \neq i$).

We verify that the product measure $\pi = \bigotimes_i \pi_i$ is stationary. For any state $\mathbf{b} \in S$:

$$\begin{aligned} \sum_{\mathbf{a} \in S} \pi(\mathbf{a}) T_{\mathbf{a}, \mathbf{b}}^{\text{intro}} &= \sum_{\mathbf{a}} \prod_k \pi_k(a_k) \cdot \frac{1}{N} \sum_{i=1}^N \pi_i(b_i) \prod_{j \neq i} \delta_{a_j, b_j} \\ &= \frac{1}{N} \sum_{i=1}^N \pi_i(b_i) \sum_{\mathbf{a}} \prod_k \pi_k(a_k) \prod_{j \neq i} \delta_{a_j, b_j} \\ &= \frac{1}{N} \sum_{i=1}^N \pi_i(b_i) \prod_{j \neq i} \pi_j(b_j) \underbrace{\sum_{a_i} \pi_i(a_i)}_{=1}, \end{aligned}$$

where in the last step the Kronecker deltas pin $a_j = b_j$ for each $j \neq i$, leaving only the sum over a_i , which equals one by normalisation. Therefore

$$\sum_{\mathbf{a}} \pi(\mathbf{a}) T_{\mathbf{a}, \mathbf{b}}^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \prod_k \pi_k(b_k) = \prod_k \pi_k(b_k) = \pi(\mathbf{b}).$$

Since T^{intro} is ergodic (by $\mu > 0$), this product measure is the unique stationary distribution. $\square$

Corollary 5 (Exact cooperation probability). *The long-run cooperation probability is*

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1 - 2\mu}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu \right]. \quad (5)$$

This is exact: no asymptotic or weak-selection approximation is made.

Proof. By Theorem 4 the stationary distribution is $\pi = \bigotimes_i \pi_i$ with $\pi_i(C) = p_i$. Since the marginals are independent, $\mathbb{E}_\pi[a_i] = p_i$ for each i , and therefore

$$p_C^{\text{intro}} = \mathbb{E}_\pi[s_{\mathbf{a}}] = \mathbb{E}_\pi \left[\frac{1}{N} \sum_{i=1}^N a_i \right] = \frac{1}{N} \sum_{i=1}^N p_i.$$

Substituting $\phi_i(\alpha_i(1 - r_i/N)) = (1 + e^{\beta_i \alpha_i (1 - r_i/N)})^{-1}$ into (4) gives (5). $\square$

5 Structural consequences

The closed form (5) allows immediate structural conclusions about how cooperation depends on game parameters.

Corollary 6 (Player-specific cooperation thresholds). *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if $r_i > N$. A sufficient condition for $p_C^{\text{intro}} > \frac{1}{2}$ is $r_i > N$ for all i .*

Proof. Since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$:

$$p_i > \frac{1}{2} \iff \phi_i(\alpha_i(1 - r_i/N)) > \frac{1}{2} \iff \alpha_i(1 - r_i/N) < 0 \iff r_i > N.$$

The average $p_C^{\text{intro}} = \frac{1}{N} \sum_i p_i$ exceeds $\frac{1}{2}$ whenever every summand does. $\square$

Corollary 7 (Neutral drift). *If $\beta_i = 0$ for all i , then $p_C^{\text{intro}} = \frac{1}{2}$ regardless of all other parameters.*

Proof. When $\beta_i = 0$, the Fermi function satisfies $\phi_i(x) = \frac{1}{2}$ for all x . Substituting into (5):

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1 - 2\mu}{2} + \mu \right] = \frac{1 - 2\mu}{2} + \mu = \frac{1}{2}. \quad \square$$

Corollary 8 (Role of heterogeneity). *For player i with $r_i \neq N$:*

- (i) $\frac{\partial p_i}{\partial \alpha_i}$ is positive when $r_i > N$ and negative when $r_i < N$.
- (ii) $\frac{\partial p_i}{\partial \beta_i}$ is positive when $r_i > N$ and negative when $r_i < N$.

Proof. Write $p_i = (1 - 2\mu)\phi_i(\alpha_i(1 - r_i/N)) + \mu$.

Part (i). Differentiating with respect to α_i :

$$\frac{\partial p_i}{\partial \alpha_i} = (1 - 2\mu)(1 - r_i/N) \phi'_i(\alpha_i(1 - r_i/N)).$$

Since $\phi'_i(x) < 0$ for all x and $1 - 2\mu > 0$, the sign equals $-\text{sgn}(1 - r_i/N) = \text{sgn}(r_i/N - 1)$, which is positive iff $r_i > N$.

Part (ii). Write $g(t) = (1 + e^t)^{-1}$, so $\phi_i(x) = g(\beta_i x)$. Differentiating with respect to β_i :

$$\frac{\partial p_i}{\partial \beta_i} = (1 - 2\mu) \alpha_i(1 - r_i/N) g'(\beta_i \alpha_i(1 - r_i/N)).$$

Since $g' < 0$ and $\alpha_i > 0$, the sign equals $-\text{sgn}(1 - r_i/N) = \text{sgn}(r_i/N - 1)$, which is positive iff $r_i > N$. $\square$

Remark 9. Corollaries 6–8 show that each player's long-run cooperation is fully determined by their individual triple (α_i, β_i, r_i) and the group size N . Players with $r_i > N$ cooperate more than half the time, and increasing α_i or β_i raises their cooperation probability further; players with $r_i < N$ show the opposite sensitivity. The parameters of other players have no effect on player i 's marginal cooperation probability.

6 Conclusion

We have proved that introspection dynamics on the heterogeneous public goods game is exactly solvable. The linear payoff structure implies that each player’s counterfactual payoff comparison is independent of all other players’ actions (Lemma 2), so each player’s 2×2 update kernel has identical rows: player i draws their next action from π_i independently of their current action. The Markov chain on $\{0, 1\}^N$ is therefore a random-scan product of N independent chains (Theorem 4), the stationary distribution is a product measure, and the cooperation probability has a closed form with N terms rather than 2^N (Corollary 5).

This closed form provides the reference point for perturbation-theoretic comparisons with other evolutionary dynamics, including the Moran process, Fermi imitation, aspiration dynamics, and introspective imitation. Non-asymptotic bounds on the resulting cooperation probability differences are pursued in companion work.

References

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